

Muhammad Achad Ansori

R&D & Robotics Engineer

achadansori@gmail.com | achadansori.com

github.com/achadansori | linkedin.com/in/achadansori

Experience

Research and Development Engineer (Contract)

Feb 2026 - Present | PT Aldzama, Gresik, Indonesia

- Researching and developing robotic systems for industrial demolition applications
- Developing industrial robotics systems; technical project details are confidential
- Collaborating with cross-functional teams to transition prototypes into field-ready solutions

Demolition Robot Technician (Contract)

Sep 2025 - Jan 2026 | PT Aldzama, Gresik, Indonesia

- Worked on maintenance and technical support for demolition robots
- Supported equipment troubleshooting and improvement work
- Subsequently moved into the R&D Engineer role

Electronics Engineer (Internship)

Dec 2024 - May 2025 | PT Venambak Kail Dipantara, Sidoarjo, Indonesia

- Redesigned PCB layout with ESP32 for FeederVE control box, handled full production cycle for 50+ units
- Developed embedded software and electronic systems for AutoVE (autonomous mobile aerator)
- Resolved electromagnetic interference issues on BoxVE system through PCB redesign

ROV Pilot and Engineer (Collaborative Project)

Jul - Aug 2024 | PT Pertamina Hulu Energi West Madura Offshore, RIG Taurus, Madura, Indonesia

- Operated ROVs for underwater hull inspections at offshore RIG, inspecting 3 RIG points at 20m depth
- Performed maintenance and troubleshooting on ROV control systems, cameras, and sensors
- Collected and analyzed visual/sensor data for hull condition assessment

Maintenance and Improvement Engineer (Internship)

Feb - Jul 2024 | PT Toshin Prima Fine Blanking (Indoprima Group), Surabaya, Indonesia

- Developed and deployed YOLOv4-based visual inspection system on Jetson Nano achieving approximately 29 FPS
- Built IoT monitoring system for 2 critical factory chiller units using Raspberry Pi with web dashboard
- Performed routine maintenance and repair of industrial fine blanking machines

Embedded Developer

Mar 2023 - Nov 2024 | Penship EEPIS Marine Robotics Team, Surabaya, Indonesia

- Built entire electrical and software systems from zero, progressing from Python+MAVLink to ROS-based autonomy stack
- Developed computer vision pipelines from HSV-based detection to deep learning (YOLO) for AUV and ASV
- Achieved 3rd Place National in KRBAI and 2nd Place Regional

General Manager

Mar 2023 - Dec 2023 | Penship EEPIS Marine Robotics Team, Surabaya, Indonesia

- Led 15-member team with 4 robot platforms, establishing weekly meeting culture from scratch
- Aligned hardware, software, and mechanical workstreams to competition deadlines

Hardware Specialist

Mar 2022 - Feb 2023 | Panship EEPIS Marine Robotics Team, Surabaya, Indonesia

- Engineered electrical systems for Autonomous Semi Submarine Vehicle, propulsion, power management, and waterproofing
- Designed custom PCBs aligned with mechanical specifications for underwater operation
- Supported software team with microcontroller programming for mission-based autonomous movements

Technical Skills

Programming

- Python, C/C++, JavaScript, Bash

Embedded Systems

- STM32, ESP32, Arduino, Raspberry Pi, Jetson Nano, MAVLink, PID Control, UART/SPI/I2C, RTOS

Robotics & Autonomy

- ROS, OpenCV, YOLO, TensorFlow, PyTorch, Sensor Integration, Navigation, Linux

Electrical

- PCB Design, Soldering & Prototyping, Wiring & Panel Assembly, Motor Drivers, Power Distribution, Oscilloscope & Multimeter

Internet of Things

- IoT Monitoring Systems, Web Dashboard, MQTT, Flutter, React, Git

Education

Bachelor of Applied Science, Electronics Engineering

2021 - 2025 | Politeknik Elektronika Negeri Surabaya (PENS), Indonesia

- Advisor: Arif Irwansyah
- Thesis: Autonomous Surface Vehicle with Collision Avoidance for Offshore Energy Exploration

Machine Learning Cohort

Aug - Dec 2024 | Bangkit Academy led by Google, Tokopedia, Gojek & Traveloka

- Completed 5-month intensive program in Machine Learning, Deep Learning, TensorFlow, and Computer Vision
- Led capstone project "InSight", YOLO-based defect detection system for SME quality control on Jetson Nano

Honors and Awards

2026

- Most Active and Best Design, Kaizen Event (PT Aldzama)

2025

- 2nd Place, Final Project Competition Applied Electronics Engineering (PENS)

2024

- Funded Proposal, Pekan Kreativitas Mahasiswa (PKM), Ministry of Education
- 3rd Place National, Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia
- 2nd Place Region 2, Underwater Robotics Competition (KRBAI), Kontes Robot Indonesia
- Finalist, Autonomous Surface Vehicle Competition (National Unmanned Fast Boat Contest)

2023

- Finalist, Autonomous Tourism Surface Vehicle Competition (National Unmanned Fast Boat Contest)
- Finalist, Boat Race Competition (COMET, National Shipbuilding Institute of Surabaya)

2022

- Finalist, Autonomous Semi Submarine Vehicle Competition (National Unmanned Fast Boat Contest)
- Finalist, UI/UX Competition (UPN Veteran Jawa Timur)

Organization & Activities

Leader & Embedded Developer

Feb - Aug 2024 | PKM-KC Project "Unmanned Autonomous Surface Vehicle for Offshore Wind Turbine Surveying"

Media & Information Team Member

Feb 2022 - Jan 2024 | EBIO PENS, Surabaya, Indonesia

- Created monthly Instagram content & designed event posters
- Led publicity, decoration, and documentation team for "Peduli Desa Jilid IX"

Certifications

2024 - 2027

- Sea Survival and Offshore Emergency Procedure (PT Global Saftindo, Credential ID: GS-ADM-024-3594)

Academic Interests

Robotics and Autonomous Systems

- Autonomous Vehicles (AUV, ASV, ROV)
- Obstacle Avoidance and Navigation

Embedded Artificial Intelligence

- Edge Computing on Jetson Nano
- YOLO-based Object Detection

Internet of Things (IoT)

- IoT Monitoring Systems
- Integration with Industrial Automation

Other Interests

Trail Running, Hiking, Reading